

The log-shift operator:

$$\int_0^\infty t^{\nu-1} (c - \log t)^p e^{-\beta t + \beta a \sqrt{t}} dt = (c - \partial_\nu)^p S_\nu(a)$$

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Abstract

Unit 14 of the grammar-paper §4 autoformalisation: the log-shift operator identity `lem:log_shift`, $\int_0^\infty t^{\nu-1} (c - \log t)^p e^{-\beta t + \beta a \sqrt{t}} dt = \sum_{m=0}^p \binom{p}{m} c^{p-m} (-1)^m \partial_\nu^m S_\nu(a)$, i.e. $(c - \partial_\nu)^p S_\nu$ written out. A short binomial argument on top of the general- k order-derivative identity (unit 13). Zero `sorry`/`axiom`.

1 Setting

Unit 13 gave $\partial_\nu^m S_\nu(a) = \int_0^\infty (\log t)^m t^{\nu-1} e^{\dots} dt$. The log-shift lemma packages a whole *shifted* log-arithm $(c - \log t)^p$ under the integral as the polynomial $(c - \partial_\nu)^p$ in the order-derivative — the operator notation the paper uses to compactly write integrals with log factors.

2 The thing formalised

```
theorem fluctuation_log_shift ( a c : ℝ ) (p : ℕ) (h : 0 < c) (h : 0 < a) :
  ( t in Set.Ioi 0, t^(-1) * (c - Real.log t)^p * Real.exp (-*t + *a*Real.sqrt t))
  = ∑ m in Finset.range (p+1), (p.choose m : ℝ) * c^(p-m) * (-1)^m
    * iteratedDeriv m (fun l => fluctuation l a)
```

3 Proof strategy

Expand $(c - \log t)^p = (-\log t + c)^p = \sum_m (-\log t)^m c^{p-m} \binom{p}{m}$ (`add_pow`) pointwise, distribute the $t^{\nu-1} e^{\dots}$ factor through the finite sum, and turn $\int \sum$ into $\sum \int$ (`integral_finsetSum`, each summand integrable via unit 13's `log_pow_fluctuation_integrableOn`). Then `integral_const_mul` pulls the binomial constants out and `iteratedDeriv_fluctuation_order` identifies each $\int (\log t)^m t^{\nu-1} e^{\dots}$ with $\partial_\nu^m S_\nu$.

4 Roadblocks and resolutions

None of substance. `setIntegral_congr_fun` again hands an applied-lambda pointwise goal, so `beta_reduce` precedes the binomial rewrite; `neg_pow` turns $(-\log t)^m$ into $(-1)^m (\log t)^m$ and `ring` matches the per-term factors.

5 What was Mathlib, what was new

Mathlib supplied `add_pow` (binomial), `integral_finsetSum`, `integral_const_mul`, `neg_pow`. New is the assembly on top of unit 13: the signed-summand integrability (norm factorised through `abs_pow`) and the finite-sum reorganisation identifying the shifted-log integral with the order-derivative polynomial.

6 Lessons

Once $\partial_\nu^k S = \int (\log)^k \cdots$ is available, any polynomial-in-log insertion is a binomial expansion plus `integral_finsetSum` — the only care is proving each signed summand integrable (bound its norm by the $|\log|^m$ -weighted integrand) so the sum/integral swap is licensed.

7 Follow-ups

`lem:log_generating` ($\sum_p \frac{z^p}{p!} (c - \partial_\nu)^p S_\nu = e^{zc} S_{\nu-z}$) is Taylor's theorem in the order and needs real-analyticity of S_ν in ν ; `prop:LogODE` differentiates the Weber ODE k times in λ and needs $\partial_\lambda/\partial_a$ commutation. The ladder-operator commutator $[b, b^\dagger] = 1$ still wants a Weyl-algebra framework.